

Optimization Design of the Heliostat Field Based on a Monte Carlo Algorithm

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Translator's note: Faithful translation. Equation numbering, symbols, and table values follow the source as printed; evident inconsistencies are flagged in notes but not altered.

Abstract

Addressing the heliostat-field optimization-design problem, this paper first builds a shadow-and-blocking efficiency model, combined with a Monte Carlo ray-tracing method, to solve the shadow-blocking efficiency; from the solar altitude angle, solar azimuth angle, and the direct normal irradiance, the cosine efficiency and the field's output thermal power are obtained. Second, a nonlinear objective-programming model is established with the per-unit-area heliostat output thermal power as the maximization objective. Within a given range of adjacent-heliostat spacing, the number of heliostats on each ring and the inter-heliostat spacing are found; the heliostat coordinates are determined by Campo's arrangement method. Using the shadow-blocking efficiency model, when the rated annual average output thermal power reaches 60 MW, the absorption-tower position is (0, 0, 80), heliostat size is 6 m × 6 m, install height 4 m, total number 2475, and total field area 89,100 m². Campo's method simplifies the determination of specific heliostat coordinates. Finally, keeping the rated power unchanged, a heliostat size of 7 m × 7 m and install height 4 m are designed to reach a larger per-unit-area output thermal power.

Keywords: Heliostat Field, Shadow-Blocking Theory, Nonlinear Objective Programming, Monte Carlo and Ray Tracing

1. Introduction

Tower solar thermal (CSP) power generation is a low-carbon, environmentally friendly clean-energy technology that China has promoted to meet its “carbon-peak” and “carbon-neutral” goals. On one hand, the system uses solar energy to generate power, reducing CO₂ and other greenhouse-gas emissions and easing air and water pollution; on the other, as a new power system built around new energy, it can use solar energy efficiently and generate power continuously [1], making it a renewable technology of great potential. This paper uses Monte Carlo ray tracing to model the shadow area produced by the heliostats and, under a given rated power, establishes a nonlinear objective-programming model that maximizes the per-unit-area output thermal power, thereby completing the optimization design. The study analyzes the 2023 CUMCM Problem A.

2. Model Establishment and Solution

2.1. Shadow-and-Blocking Efficiency Model

Sunlight losses in the field arise mainly in three ways: 1) a front heliostat blocks sunlight bound for a rear heliostat; 2) the light reflected by a rear heliostat is blocked by a front heliostat; 3) the tower casts a shadow that blocks some heliostats.

A heliostat-a mirror frame (X_a, Y_a, Z_a), a ground frame (X_d, Y_d, Z_d), and a heliostat-b mirror frame (X_b, Y_b, Z_b) are established. To judge whether heliostat a's reflected ray falls onto heliostat b, its coordinate value K_2 in b's frame is determined; the inter-heliostat sunlight path is shown in Figure 1.

[Figure 1. Diagram of the sun-ray path between heliostats — schematic in the original.]

For a heliostat E whose center has ground-frame coordinates ($X_{e,d}, Y_{e,d}, Z_{e,d}$), its azimuth within the field is:

$$\vartheta_f = \arccos(X_{e,d} / \sqrt{X_{e,d}^2 + Y_{e,d}^2}), \text{ if } Y_{e,d} \geq 0$$

$$\vartheta_f = \arctan(Y_{e,d} / X_{e,d}) + \pi, \text{ if } Y_{e,d} < 0$$

Let θ be the angle between the incident sunlight and the normal of heliostat b. Its relation to the solar altitude angle α_s [2] and solar azimuth angle γ_s [3] is:

$$\cos \theta = (\sqrt{2} / 2) \cdot [\sin \alpha_s \cos \theta - \cos(\vartheta_f - \gamma_s) \cdot \cos \alpha_s \sin \theta + 1]^{(1/2)}$$

The pitch and azimuth of the heliostat's central normal are:

$$F_y = \arctan((\sin \alpha_s + \cos \theta) / (2 \cos \vartheta))$$

$$F_w = \arctan((\sin \vartheta_f \sin \theta - \sin \gamma_s \cos \alpha_s) / (\cos \vartheta_f \sin \theta - \cos \gamma_s \cos \alpha_s)) \quad [4]$$

For an incidence point $K_1 = (x_a, y_a)$ in a's frame, coordinate transforms decide whether the reflected ray has an incidence point K_2 in b's frame. Writing the three axes of a's frame as ground-frame vectors (r_x, s_x, t_x), (r_y, s_y, t_y), (r_z, s_z, t_z), the transformation matrix is Eq. (2.1) [5]:

$$G = [r_x r_y r_z; s_x s_y s_z; t_x t_y t_z] \quad (2.1)$$

$$= [-\sin F_w \cos F_y, -\sin F_w \cos F_y, \cos F_w \cos F_y; \cos F_y, -\sin F_w \sin F_y, \cos F_w \sin F_y; 0, \cos F_w, \sin F_w]$$

With the cone of light expressed as P_j in the mirror frame and $P_o = G \cdot P_j$ in the ground frame, $K_2 = (x_b, y_b)$ is obtained in five steps: (1) transform a point K_1 of heliostat a into the ground frame, giving K_1' ; (2) transform K_1' into heliostat b's frame, giving K_1'' ; (3) transform the ground-frame ray vector P_o into b's frame, giving P_j ; (4) in b's frame compute the ray-mirror intersection from $K_1'' = (x_{a''}, y_{a''}, z_{a''})$ and $P_j = (x_o, y_o, z_o)$:

$$K_2 = (x_b = (z_o x_{a''} - x_o z_{a''}) / z_o, y_b = (z_o y_{a''} - y_o z_{a''}) / z_o, 0)$$

(5) judge whether K_2 falls within heliostat b's plane.

2.2. Atmospheric Transmittance

Let d_{HR} be the distance from the heliostat center to the receiver center. The atmospheric transmittance is [6]:

$$\eta_{at} = 0.99321 - 0.0001176 \cdot d_{HR} + 1.97 \times 10^{-8} \cdot d_{HR}^2 \quad (d_{HR} \leq 1000) \quad (2.2)$$

2.3. Cosine Efficiency

Construct the heliostat's reflected-sunlight diagram (Figure 2): receiver center A, heliostat center O_z, incident ray BO_z, reflected ray O_zA, and perpendiculars EF and BC to the xoy-plane with feet F and C. For the i-th heliostat with incidence angle θ_i , the cosine efficiency is [7]:

$$\eta_{cos} = [d_{HR} + \sin\alpha_s \cdot h_a + x_z \cos\alpha_s \sin\gamma_s + y_z \cos\alpha_s \sin\gamma_s]^{(1/2)} / (2 d_{HR})^{(1/2)} \quad (2.3)$$

[Figure 2. Three-dimensional representation of sunlight reflected by a heliostat — schematic in the original.]

2.4. Optical Efficiency of a Single Heliostat

The optical efficiency of a single heliostat is $\eta = \eta_{sb} \cdot \eta_{cos} \cdot \eta_{at} \cdot \eta_{trunc} \cdot \eta_{ref}$, where the shadow-blocking efficiency η_{sb} , cosine efficiency η_{cos} , and atmospheric transmittance η_{at} are given above; the literature reports the receiver truncation efficiency η_{trunc} at 60%–90% [8], and the mirror reflectivity is $\eta_{ref} = 0.92$.

Translator's note: The truncation efficiency is not modeled here; it is simply taken from the literature as 60–90%. This is the weakest link in the method, since the cone-beam “spillage” past the receiver is exactly the loss the contest emphasizes.

2.5. Direct Normal Irradiance

The direct normal irradiance [9] is:

$$DNI = G_o \cdot [a + b \cdot \exp(-c / \sin\alpha_s)]$$

$$a = 0.4237 - 0.008216(6-H)^2, \quad b = 0.5055 + 0.00595(6.5-H)^2, \quad c = 0.2711 + 0.01858(2.5-H)^2$$

where $G_o = 1.366 \text{ kW/m}^2$ is the solar constant and H is the local altitude, 3000 m.

2.6. Field Annual Average Output Thermal Power

Each heliostat's collecting area A_i corresponds to a 6 m × 6 m mirror; with each heliostat's optical efficiency η_i , the DNI, and a total of $N = 1745$ heliostats, the field output thermal power is $E_{field} = DNI \cdot \sum_{i=1..N} A_i \cdot \eta_i$. From this the average optical efficiency $\bar{\eta}$, average cosine efficiency $\bar{\eta}_{cos}$, average shadow-blocking efficiency $\bar{\eta}_{sb}$, average truncation efficiency $\bar{\eta}_{trunc}$, and per-unit-area average output power $E_{field} / (N \times A)$ are obtained (Tables 1 and 2).

Table 1. Average optical efficiency and output power on the 21st of each month (Problem 1).

Date	$\bar{\eta}$ (optical)	$\bar{\eta}_{cos}$	$\bar{\eta}_{sb}$	$\bar{\eta}_{trunc}$	$E_{field}/(N \times A)$ (kW/m ²)
Jan 21	0.543440747	0.790895784	0.88722	0.8723459	0.493146841
Feb 21	0.55498721	0.806084862	0.87231	0.8890342	0.518939745
Mar 21	0.567908242	0.81509369	0.87992	0.8918967	0.564501479
Apr 21	0.577014059	0.818542441	0.89003	0.892129	0.593527476
May 21	0.590836903	0.815817555	0.90402	0.9023679	0.616706666
Jun 21	0.596222479	0.806293553	0.92301	0.9023933	0.625373899
Jul 21	0.601603651	0.791690147	0.93289	0.9175123	0.628606574

Date	$\bar{\eta}$ (optical)	$\bar{\eta}_{\cos}$	$\bar{\eta}_{sb}$	$\bar{\eta}_{trunc}$	$E_{\text{field}}/(N \times A)$ (kW/m ²)
Aug 21	0.585272892	0.776101811	0.92331	0.9199819	0.602248936
Sep 21	0.572436585	0.765626719	0.91678	0.91861235	0.570268117
Oct 21	0.554517146	0.762241825	0.90203	0.90842344	0.520380316
Nov 21	0.544281187	0.765515871	0.89133	0.89849923	0.471132767
Dec 21	0.537198117	0.775443951	0.88544	0.8812762	0.444665057

Table 2. Annual average optical efficiency and output power (Problem 1).

Annual avg optical eff.	Annual avg cosine eff.	Annual avg shadow-blocking eff.	Annual avg truncation eff.	Annual avg output (MW)	Per-unit-area annual avg output (kW/m ²)
0.568809935	0.790779018	0.900690833	0.899539368	34.81012136	0.554124823

2.7. Campo Arrangement Method

With the absorption tower at the center, the field is built as a concentric-circle model in which each heliostat's footprint is treated as a circle. Following Campo's method, the circular field is arranged with circumferential uniformity under the principle that no two heliostats mechanically collide (Figure 3). Let the heliostat positions be (X, Y) , with (x_i, y_i) the ground-frame coordinates of the i -th heliostat, $(x_{b,i}, y_{b,i})$ the reflected-point coordinates in heliostat b , and $x_{i,j}$ the i -th point on the j -th ring. The nonlinear-programming model is Eq. (2.4):

[Figure 3. Heliostat-field model: concentric rings about the tower — schematic in the original.]

$$\text{maximize } E_{\text{field}} = DNI \cdot \sum_{i=1..N} A_i \cdot \eta_i \quad (2.4)$$

subject to:

$$DNI = G_o [a + b \cdot \exp(-c / \sin \alpha_s)]$$

$$\eta_i = \eta_{sb,i} \cdot \eta_{\cos,i} \cdot \eta_{at,i} \cdot \eta_{trunc,i} \cdot \eta_{ref,i}$$

$$\eta_{at,i} = 0.99321 - 0.0001176 \cdot d_{HR} + 1.79 \times 10^{-8} \cdot d_{HR}^2 \quad (d_{HR} \leq 1000)$$

$$\eta_{\cos,i} = \cos \vartheta_i = [d_{HR} + h_a \sin \alpha_s + x_i \cos \alpha_s \sin \gamma_s + y_i \sin \alpha_s \sin \gamma_s]^{1/2} / (2 d_{HR})^{1/2}$$

$$\eta_{sb,i} = (x_{b,i} - 3)(y_{b,i} - 3) / 36$$

$$100 < |x_i| < 350, \quad 100 < |y_i| < 350, \quad 11 < a < 14$$

$$\sqrt{(x_{(i+1,j)} - x_{(i,j)})^2 + (y_{(i+1,j)} - y_{(i,j)})^2} > a$$

$$\sqrt{(x_{(i,j)} - x_{(i,j+1)})^2 + (y_{(i,j)} - y_{(i,j+1)})^2} > a$$

$$r_j = 106 \quad (j = 1); \quad r_j = 106 + a \cdot j \quad (j > 1), \quad \text{with } 100 < |r_j| < 350$$

Translator's note: Two source inconsistencies are preserved above: the η_{at} coefficient is printed as 1.79×10^{-8} in Eq. (2.4) but 1.97×10^{-8} in Eq. (2.2); and the keep-out constraint is written as a square box $(100 < |x| < 350 \text{ and } 100 < |y| < 350)$ rather than the problem's true annulus $100 < \sqrt{x^2 + y^2} < 350$.

Using Python, the inter-heliostat spacing is swept from 11 m to 14 m in 0.1 m steps. With r the radius of a heliostat center on the field ring and R the field radius, the number of heliostats per ring is $n = 360 / [2 \cdot \arcsin(r/R)]$. The arrangements for different spacings are illustrated in Figure 4. The optimal adjacent

spacing is 12 m: at the rated annual average output of 60 MW, the tower position is (0, 0, 80), heliostat size 6 m × 6 m, install height 4 m, total number 2475, and total field area 89,100 m².

[Figure 4. Heliostat arrangements for spacings 11.5, 12, 12.5, 13, 13.5, and 14 m — six plots in the original.]

2.8. Achieving a Larger Per-Unit-Area Output

Keeping the rated power fixed, the size and install height are designed for a larger per-unit-area output. The mirror size is increased from 6 m × 6 m to 7 m × 7 m [10], the largest size that still fits within the spacing; the install height then increases progressively from the inner to the outer rings, over the range 3.5–6 m. To find the number of heliostats per ring, a MATLAB search [11] counts the heliostats within each ring's radius range and subtracts successively; the install height is divided into 20 equal parts, the per-ring values are tabulated in Excel, and these, together with an install-height relation, are fed into the Python code. Results appear in Tables 3, 4, and 5.

Table 3. Average optical efficiency and output power on the 21st of each month after optimization (7×7 design).

Date	$\bar{\eta}$ (optical)	$\bar{\eta}_{\cos}$	$\bar{\eta}_{sb}$	$\bar{\eta}_{trunc}$	$E_{\text{field}}/(N \times A)$ (kW/m ²)
Jan 21	0.523745568	0.763128059	0.88722	0.8723459	0.475274395
Feb 21	0.532102309	0.773754046	0.87231	0.8890342	0.49754126
Mar 21	0.549914813	0.790195917	0.87992	0.8918967	0.546615989
Apr 21	0.56768292	0.806251682	0.89003	0.892129	0.583929291
May 21	0.590019553	0.815646237	0.90402	0.9023679	0.615853529
Jun 21	0.605040133	0.819179402	0.92301	0.9023933	0.634622679
Jul 21	0.619645784	0.816391126	0.93289	0.9175123	0.647458526
Aug 21	0.607461142	0.806471063	0.92331	0.9199819	0.625080764
Sep 21	0.590742135	0.791038547	0.91678	0.91861235	0.588504323
Oct 21	0.562741657	0.774456199	0.90203	0.90842344	0.528098515
Nov 21	0.542031167	0.763247051	0.89133	0.89849923	0.469185138
Dec 21	0.525610674	0.759609008	0.88544	0.8812762	0.435073565

Table 4. Annual average optical efficiency and output power (7×7 design).

Annual avg optical eff.	Annual avg cosine eff.	Annual avg shadow-blocking eff.	Annual avg truncation eff.	Annual avg output (MW)	Per-unit-area annual avg output (kW/m ²)
0.568061488	0.789947361	0.900690833	0.899539368	67178.64878	0.553936498

Translator's note: The "annual average output thermal power" is printed as 67178.64878 MW, which is impossible for a ≈60 MW plant. The value is actually in kW: 0.553936498 kW/m² × 121,275 m² ≈ 67,178 kW ≈ 67.2 MW. So this 7×7 design also overshoots the 60 MW rating. Reproduced as printed.

Table 5. Design parameters (7×7 design).

Absorption-tower position	Heliostat size (W × H)	Install height (m)	Total number	Total area (m ²)
(0, 0, 80)	7 m × 7 m	4	2475	121,275

3. Conclusion

For the heliostat-field optimization-design problem, this paper establishes a shadow-and-blocking efficiency model and a nonlinear objective-programming model that maximizes the per-unit-area output thermal power. With the heliostat size fixed, Tables 1 and 2 give the field's annual average optical efficiency, annual average output thermal power, and per-unit-area annual average output. At a rated annual average output of 60 MW, the optimal adjacent-heliostat spacing is 12 m; the tower position is (0, 0, 80), heliostat size 6 m × 6 m, install height 4 m, total number 2475, and total field area 89,100 m². Keeping the rated power fixed and maximizing the per-unit-area annual average output yields the design parameters in Table 5.

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